Semester: 5th
Subject Name:- Compiler **& Code:- CM30006**
Branch (s): - CSE

AUTUMN **MID SEMESTER EXAMINATION-2024**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name: Compiler
[Subject Code: **CM30006**]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

a) Consider the following C-programme. Find the number of tokens in the output of the following program if that is passed as a string to a lexical analyzer.

```
#include <stdio.h>
int main()
{
    char *p[ ] = {"KIMS MED", "KIIT UNIV","KISS"};
    char **q[ ] = {p , p + 1, p + 2};
    char ***r=q;
    printf("I am in %s", ** ++ r);
    return 0;
}
```

b) Consider the following grammar:

S-> A & S | S $ A

A-> B @ B

A-> id

B-> id

Arrange the precedence of the operators &, $, @ in increasing order.

c) Explain shift-reduce conflict with an example.

d) Find FIRST() & FOLLOW() of the following grammar.

S → AB|a

A → BAb | ε

B→ c | ε

e) Eliminate left recursion from the following grammar.

S→ Sa | SAb | a | b

A→ Aa | Ab | a | b

2. a) Show the output for each phase for the expression S = a + b * c + 20 with a neat and clean diagram of translation of the given statement (where the variables a ,b,c are of integer type and S is of floating type). [3 Marks]

   b) Analyze the role of Lexical and syntax analyzer with the possible error recovery actions. [2 Marks]

3. a) Remove left recursion from the following grammar:

   S -> Xa / Sa / b

   X -> Xc / Se / d [2.5 Marks]

   b) Left factor the following grammar:

   S-> abcd | abce | abf | ag

   S->aA | abAa | abcA | abA

   A-> ab| b [2.5 Marks]

4. a) Construct the FIRST() and FOLLOW() sets of the following grammar:

   A -> B % C

   B -> C + A | a

   C -> A # C | b

   Here, the capital letter indicates non-terminals. [2.5 Marks]

   b) S -> (X) | ab
   X -> S@Y
   Y -> SY) | ε
   Here, the capital letter indicates non-terminals. Check the above grammar is in LL(1) or not. Justify your answer. [2.5 Marks]

5. a) S → Aa | bAc | Bc | bBa

   A → d

   B→d

   Check whether this grammar is LR(0) or not. [2.5 Marks]

   b) Write down the procedures for the recursive descent parser for the given grammar:
   E -> 0E*A / E+A / E%0
   A -> 1A01 / 1
   Compare between predictive recursive and predictive non-recursive descent parser. Which one is better justify your answer. [2.5 marks]

*** Best of Luck ***